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Student Poster Presenter Titles and Abstracts

Angela M. Grant Student Poster Session: Thursday, June 27 at 06:00 – 08:00 pm

Conference for African American Researchers in the Mathematical Sciences

Department of Mathematics, Tufts University, June 27 – June 28, 2024

Felix Ackon

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Analysis of Texas's Power Grid: Quantifying Localized Carbon Emissions

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This study introduces a novel methodology for co-determining explicit and implicit carbon emissions of different substations within the Texas grid. By incorporating random renewable energy generation (including wind and solar power) and random load patterns across the state into a large-scale grid, our method provides a more frequent and realistic assessment of carbon emissions associated with electricity generation.

Regarding carbon pollution, blame is assigned to the dirty fossil fuel power plants, with the coal plants accruing the most blame and the natural gas plants spared slightly due to their lower emissions. Thus, policymakers simply propose phasing out coal plants with a cleaner source to have the most significant impact on carbon reduction. However, omitting the demand and transmission side of the problem leads to inadequate supply capacity – intensified by intermittency – and suboptimal capacity planning decisions.

Our proposed quantity, the Carbon Export Score, leverages the hourly dispatch and load across various locations to quantify the emissions from electricity transmission within an area for various dispatch scenarios and to assign blame to importers and producers. Results indicate that simply phasing out coal plants with renewables results in a modest improvement in carbon emissions if the new generators are not located near areas of demand concentration, especially during peak demand hours. The Carbon Export Score highlights the importance of spatial and temporal factors in assessing the environmental impact of power generation, offering a robust tool for advancing towards a more sustainable and low-carbon energy future to grid operators.

Emmanuel Adara
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Using the Moment Method to Solve the Stochastic Reaction-Diffusion Master Equation for Biochemical Systems

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The Reaction-Diffusion Master Equation (RDME) is a mathematical framework used to model the spatiotemporal dynamics of interacting particles undergoing reactions and diffusions in a confined space. It is particularly relevant in the study of biological systems, such as the behavior of molecules within cells or the dynamics of ecological populations. Mathematically, the stochastic RDME ultimately leads to a set of ordinary differential equations that describe how the probability distribution of particles evolves over time and space. The key components include reaction terms, accounting for particle interactions, and diffusion terms, reflecting the random motion of particles through space.

In this research, we apply the compartment-based modeling approach in formulating the system and using the moment method as an efficient alternative to computing a full statistical measure such as the mean number of molecules or population of the reaction-diffusion master equation (RDME) in comparison to other well-known approach such as the stochastic simulation algorithm (SSA).

Bernard Akwei

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Minimal spacing of Eigenvalues on Fractals

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On the unit interval (I), and the Sierpinski Gasket (SG), the spectral decimation function of the Laplacian has similar properties that result in positive minimum spacing of eigenvalues.

Other fractals, for example the level-3 Sierpinski Gasket (SG_3), may not necessarily enjoy these properties.

Our goal is to obtain an easy and sufficient criterion for positive infimum spacing of eigenvalues in the spectrum based on the properties of the spectral decimation function for the appropriate fractal.

We also give a sufficient condition for zero infimum spacing of eigenvalues.

Leila Alston and Amara Nwokoye

Hampton University and University of Maryland

Generating Functions in the Study of Random Walks and Combinatorial Path Structures

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This project dives into three unique sequences. The Catalan numbers produce the number of ways we can draw trees with a certain amount of edges. For each step taken, the path along the Catalan numbers will either take a path going up or going down. The Motzkin numbers produce the number of ways we are able to draw edges across a circle without any of them intersecting each other. The options for each step are similar to the Catalan numbers, with the exception to including one full step to the right (1,0). Finally, we have the Schröder numbers which count the number of Lattice paths from the point of origin to a fixed point (n,n). The steps taken are similar to the Motzkin path with the exception of the horizontal step taking two of the right in one instance (2,0) rather than a single step. With all three, we are able to produce generating functions for each to better characterize them in their own respect.

This research is made possible by the generous support of the National Security Agency (Grant number H98230-24-1-0025), National Science Foundation (Grant number 2243912), the American Mathematical Society, Georgetown University and American University through the SPIRAL REU and SPATIAL-Stats REU.

Elisha Amadasu

DePauw University

Zermelo's Theorem: How to Reach a Standard of Perfect Play in Chess

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Zermelo's theorem establishes that in any two player zero-sum game with perfect information (without the element of chance), either one side can force a win regardless of how the other side plays, or both sides can force a draw (if allowed in the rules). This theorem encouraged me to see how I could better my chess to see if I could exploit the existence of this perfect standard.

I considered how chess engines play, as they are currently the strongest at playing chess. I understood that they often look 60-70 moves deep from the opening stage of the game, which leads to the endgame. I decided to mimic this method by finding a statistical correlation between opening moves and the type of endgames that may occur from them.

Blossom Anyanwu and Loren Nix

University of Maryland and University of Texas

Exploring Rank-Based Metrics with Spearman's ρ

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Functional Data Analysis (FDA) is a field of statistics that explores data where observations are continuous functions as opposed to single values. These functions arise from diverse areas like EEG recordings, sensor measurements, data collected over intervals of time, and even climate data. Developing reliable methods for analyzing this type of data that extends beyond traditional statistics is crucial.

We extend Spearman's rank correlation for analyzing trends within functional data using the doubly ranked framework. Spearman's rank correlation coefficient (ρ), a nonparametric measure, assesses the monotonic relationship between two variables. Unlike other correlation tests such as Pearson's Coefficient, Spearman's does not require any base assumptions to be made regarding the data's distribution. By assessing the rank-based relationships between functions, Spearman's rank allows us to identify relationships between functional variables.

We implement this non-parametric approach to various functional data sets in the statistical software R. We illustrated our method by comparing sea surface temperatures of the ElNino_ERSST_region_1and2 and ElNino_ERSST_region_3 data sets within the 'Rainbow' R package. In the future we hope to expand this process to multiple data sets to examine the strengths and weaknesses of this metric.

This research is made possible by the generous support of the National Security Agency (Grant number H98230-24-1-0025), National Science Foundation (Grant number 2243912), the American Mathematical Society, Georgetown University and American University through the SPIRAL REU and SPATIAL-Stats REU.

Sébastien Bernard

Ridgefield High School

Object Detection in Self-Driving Cars

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Object detection is a critical task that ensures safe and efficient navigation in self-driving cars. One of the main components of this task consists of determining what the objects are and assigning them to predefined categories based on their features. This project's objective is to develop a deep learning system capable of distinguishing images of objects on the road. To assess the effectiveness of our system, we will benchmark its performance against a state-of-the-art expert model.

Terra Colvin, Jr.

Tufts University

Universality of Clifford+T circuits using approximations to $R_Z(\theta)$

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As we transition from the noisy intermediate-scale quantum (NISQ) era into the early fault-tolerant quantum computing era and beyond, the need for quantum error correcting codes will necessitate a shift from shallow, parameterized circuits to deeper, error-correctable circuits.

Clifford+T is a popular universal gateset, but implementing the T operation often requires considerable device resources. Minimizing the number of T gates required to implement an arbitrary unitary is an area of active research. This work introduces a popular algorithm for discrete approximations to the parameterized $R_Z(\theta)$ gate using Clifford+T gates. Relevant topics including matrix norms, universality and unitary approximations, and the relation between T-count and approximation precision are also presented. Finally, we show how to move beyond single-qubit unitaries and synthesize error-correctable circuits with any number of qubits.

The goal of this work is to provide a Python implementation of the Haskell package *gridsynth* and to make the underlying mathematics accessible to more of the quantum computing community. This will facilitate easier integration into existing quantum computing SDKs such as Cirq and Qiskit.

Aquilah Daughtery and Rishi Siddharth

Xavier University of Louisiana, American University/LSE

Daily Stock Optimization Tool Based on Political and General Sentiment

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Investing in profitable stocks can be challenging for novices. Our research supports these investors with insights from real-time data analysis using yfinance. We categorized stocks into five sectors and used VADER for analysis. Stocks were evaluated on metrics like Dividends, EPS, Market Cap, Revenue, P/E Ratio, and Political Sentiment, with each metric weighted by time period. Sentiment analysis impacts short-term volatility. Daily stock updates are at 12 am EST using Python, HTML, and SQL.

This research is made possible by the generous support of the National Security Agency (Grant number H98230-24-1-0025), National Science Foundation (Grant number 2243912), the American Mathematical Society, Georgetown University and American University through the SPIRAL REU and SPATIAL-Stats REU.

Orville Dillon, Jr.

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CSV to 3D: Demonstration of Turning Numbers into Images

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This is a demonstration of using Python to produce a 3D image from the CAARMS attendee survey dataset, sanitized, and used as an input via Comma-Separated Value(CSV) file.

The walkthrough covers uses of Bash, and Python, running a one-liner from each, directing results to files that will later be used in the Python source code to create the image(s).

The survey dataset was derived from a questionnaire provided to all registered attendees of CAARMS 2021. The surveyed were requested to check the boxes of mathematical disciplines that interest them.

The options for data structures will be shown, as they each have overlapping capabilities, which lends opportunity to achieving the same graph/image through use of different tools offered by high-level programming languages such as Python, Java, C, etc.

Zachary Duah

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Morita Equivalence of Inverse Hulls of Markov Shifts

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In this research, we characterize the Morita equivalence of inverse hulls of a class of inverse semigroups associated with Markov shifts. In particular, we demonstrate how to construct a labeled graph that is a complete invariant of the Morita equivalence classes of inverse hulls of semigroups associated with Markov shifts over finite alphabets.

Amara Rebecca Eze

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An Efficient Fixed-Point Algorithm with Inertial Step for Solving Variational Inclusion Problems with Applications

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Developing iterative schemes for variational inclusion problems (VIP) is a dynamic research field. This presentation provides an overview of current VIP resolution algorithms. We introduce an efficient Mann-type algorithm with inertial extrapolation terms and self-adaptive step sizes, enhancing convergence without stringent conditions. The presented version is flexible, with easily implementable criteria for the inertial factor and relaxation parameter. Numerical experiments illustrate the practical implementation and performance of the proposed algorithm.

Jessica Ezemba

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Shape of Uncertainty: Variational Finite Element Analysis for Aleatory Approximation Using 3D Geometry Representation

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New products are usually created by enhancing the functionality and design of existing ones by redesigning components and their specifications to reduce design lead times and costs and to improve product quality. A common way to improve the robustness of redesigned products is to conduct Finite Element Analysis (FEA) of predefined conditions. However, this deterministic FEA approach is often conducted in late development phases, leading to products with a high safety factor or overly optimistic designs prone to failure as uncertainties associated with future excitations and system modeling occur.

One way to model these uncertainties is by representing them as a bathtub curve comprising three stages: the initial stage with a decreasing failure rate, the middle stage with an approximately constant failure rate, and the final stage with an increasing failure rate. Uncertainties in this middle stage are studied under uncertainty quantification analysis as aleatoric uncertainty which characterizes intrinsic variation of the system caused by model input parameters, which can lead to an unpredictable outcome variation.

Our method addresses aleatoric uncertainty by using variational finite element analysis (FEA) to model stress distribution on different geometries. Using Monte Carlo Simulation to vary environment load size, frequency, and magnitude, and encoder-decoder graph model our method predicts the likely stress distribution of an unseen 3D model based on designer inputs about the belief of the environment. The results show that stress distribution can be predicted by learning the mapping between geometry and environmental load conditions.

Trajan Hammonds

Princeton University

Uniform Asymptotics for Relative Characters on GL2

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The asymptotics of relative characters for real Lie groups were studied for GanGross-Prasad pairs (G, H) by Nelson and Venkatesh in 2021. There, they develop the “microlocal analysis of Lie group representations”, which involves characterizing vectors by the property that they are approximate eigenvectors under the action of small subgroups. They compute the asymptotics whenever the conductor of the Rankin-Selberg L-function lies in a stable locus, i.e. away from conductor dropping. In the non-archimedean case, such approximations become exact and the resulting analysis is much simpler. This allows us to extend their results in the case of (GL2, GL1) to the situation where the conductor drops significantly. These local harmonic-analytic results may open the door to solving the subconvexity problem for L-functions in more general situations.

Ananya Ramesh and Nicholas Tukuru

Georgetown University and Boston College

Rethinking Rank-Based Metrics: Kendall's Tau for Nonparametric Functional Data

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A major challenge in functional data analysis, which provides insight from data collected over a continuous interval, is that it has limited options for rank-based correlation metrics. We develop an extension of Kendall's tau, a nonparametric measure for monotonicity, to address this problem. This adds to a growing methodology of distribution-free analysis for functional data and facilitates the extraction of new information and more robust analyses.

To extend Kendall's tau to functional data, we first rank the functions in each dataset at each measured column, or unit of time. Then, we summarize the ranks by taking the average of each row, or group. Lastly, we compare the summarized rank statistics of the first and second dataset, using Kendall's tau methodology. To illustrate this, we use two related datasets that contain monthly sea surface temperature measurements in different regions of El Niño over 69 years. Applying this function returns a metric of monotonicity that is relevant to the datasets. We discuss the importance of Kendall's tau in functional data analysis for its distribution-free approach that allows for wider application.

This research is made possible by the generous support of the National Security Agency (Grant number H98230-24-1-0025), National Science Foundation (Grant number 2243912), the American Mathematical Society, Georgetown University and American University through the SPIRAL REU and SPATIAL-Stats REU.

Jaylen Rivera and Eric Wiggins

The City College of New York and Virginia Commonwealth University

Exploring the Applications of the Catalan Numbers and their Relations to Other Trees

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This project dives into the Catalan Numbers and how they came to be. With the ways the function can be imaged, we are able to solve for a generating function C(x) to better characterize the sequence.

We visualize it through a random walk with each path going either strictly up or down for each step taken to the right. One purpose of the Catalan numbers is identifying the number of ways you draw rooted trees based on the number of edges (n).

For instance, for the nth term in the sequence, we would have n edges to work with. So, for 3 edges, we would need the third term of the sequence which is five. Therefore, for 3 edges, there are 5 different rooted trees we can draw. With that, we can form various bijections directly related to the Catalan sequence.

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Jhevon Smith

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Closure Approximations for the Erlang-A Queueing Model

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The Erlang-A model is fundamental in queueing theory. It has many practical and theoretical applications, making it widely studied by queueing theorists. In the most general sense, it models the idea of customers randomly joining a queue to obtain some service, waiting for a random amount of time, after which they either receive the service or abandon the queue all together as they grow too impatient.

This models many real-world scenarios, such as: waiting to get help on a customer service call line, waiting in a hospital emergency room to receive medical attention, driving around looking for parking, or computer programs making requests to a server before being processed or timing out. Tweaks can be made to model highly impatient customers who leave the moment they join the line and are not served, to highly patient customers who never abandon the line before getting served, to a scenario with infinite servers where every customer receives service immediately upon arrival. Such useful applications compel us to analyze this model efficiently and accurately.

Traditional methods for analyzing this queue, over some finite time horizon, include tedious analytical calculations or Monte Carlo simulations requiring hundreds of thousands of iterations. We provide an expedient alternative by developing the heuristic of *closure approximations*. This leads us to build the tools and formulas needed to analyze the transient cumulant moments of the Erlang-A queue by running a *single* simulation run of some dynamical system.

Odirachukwunma Ugwu

Morgan State University

An Efficient Iterative Method for Solving Bilevel Variational Inequality Problems in Hilbert Spaces Involving Monotone Operators.

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In this study, we investigate a class of problems known as the Bilevel Variational Inequality Problems and propose an efficient and accurate method for solving them involving monotone operators. Motivated by the application of BVIP in emerging technologies such as smart cities, renewable energy systems, and advanced transportation networks. Solving BVIPs efficiently can unlock the potential of these technologies to address pressing societal challenges such as climate change, urbanization, and sustainable development. Our proposed method for solving this class of problems consists of inertial extrapolation step technique, a self-adaptive step-size to improve the speed of convergence of the scheme and a single projection onto a half space which reduces the computational cost compared to some existing related work. Strong convergence is obtained under some relaxed conditions on the control parameters. Finally, some numerical results are presented to illustrate the effectiveness of our proposed algorithm compared to other related ones in the literature.

Kamille Watkins and Ariana Abdullah

Spelman College and George Washington University

Predictive Model for a Time-Based Stock Optimization Tool

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This project developed and analyzed a predictive model for stock portfolio recommendations using Python libraries 'yfinance', 'pandas', and 'statsmodels'. Historical stock data for tickers across various sectors (AI/ML, Entertainment, Food, Health Care, and Oil) were collected and analyzed to investigate relationships between key financial indicators and stock performance metrics such as daily returns and last month's average price. The dataset included information like price, volume, and financial ratios. Key predictors identified were PE ratio, dividends, beta, market cap, profit margins, among others. Ordinary Least Squares (OLS) regression was used to model the impact of these financial variables on daily returns and last month's average price. The results highlighted important determinants of stock performance and provided insights into the financial health and potential future returns of stocks. Visualizing correlations enhanced our understanding of the relationships between stock returns and financial indicators. This research aids in identifying stock volatility and leveraging financial data for investment decisions.

This research is made possible by the generous support of the National Security Agency (Grant number H98230-24-1-0025), National Science Foundation (Grant number 2243912), the American Mathematical Society, Georgetown University and American University through the SPIRAL REU and SPATIAL-Stats REU.

Madushi Wickramasinghe

Morgan State University

A Second Order Numerical Method for Solving Reaction Diffusion Equations

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There are different exponential time differencing (ETD) schemes for solving nonlinear differential equations such as ETD-Padé(1,1) or ETD-Crank-Nicolson scheme and ETD-Padé(0,2) scheme which use various Padé rational function approximations of the matrix exponential. This study presents an efficient second order exponential time differencing scheme which utilizes the real distinct poles discretization for the matrix exponential called ETD-RDP. This scheme allows parallel implementation. A numerical example is presented to compare the suggested scheme with two other second order schemes, ETD-Crank-Nicolson scheme and ETD-Padé(0,2) scheme.

Reuel Williams

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Neural Networks Learn Through Shape

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The prevailing goal of Persistent Homology in Topological Data Analysis is to study the shape of complex datasets by observing which homological features persist as some continuous parameter ϵ is changed. This is formalized by constructing simplicial complexes, which are triangulations of topological spaces, in which more simplices are added as ϵ increases. This then forms a nested collection of simplicial complexes in which homology classes may be born or die. The longer a class persists, the more likely it indicates some underlying shape of the data.

Persistent homology provides a promising path for neural network interpretability, as one may understand how a neural network completes a task, or how well it may generalize, based on what its shape is. In this research I reproduce and extend the results of Naitzat et al. (2020) to show that multilayer perceptrons on simple discrete classification tasks work by changing the topology of the input layer by layer. Specifically, higher dimensional homology groups become trivial while more connected components are formed.

Saba Zerefa

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Efficient Protein Structure Comparison via Amino Acid Neighborhoods

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Improvements in protein structure prediction methods have brought rapid growth in publicly available protein structures. This leads to larger databases of protein structures, which in turn demands efficient means of searching such databases. Previous work (Trinquier et al. 2022) developed SWAMPNN, a neural network that inputs the 3D coordinates of a protein pair and outputs a structural alignment. However, training neural networks is computationally expensive and is not strictly necessary for this task. We have developed a means of determining a structural alignment between a pair of proteins by considering aspects of protein structure local to each amino acid, encoded in a vector called a neighborhood. Each neighborhood included secondary structure information and relative position of amino acids contained within 15 Angstroms of a reference amino acid; such amino acids constitute the "neighbors" of the reference amino acid. By characterizing relatedness between two such neighborhoods of amino acids, we are able to create a scoring mechanism compatible with the Smith-Waterman algorithm, and thus can construct a local sequence alignment. Our method performs similarly to LDDT of structural alignments in the Dali dataset, and thus can be used for rapid protein structure comparison in protein database search tasks.

Carmel Prosper Sagbo

Carnegie Mellon University Africa

Cancer Evolutionary Process: Synthetically Generating a New Type of Cancer Using Probabilistic Graphical Models

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The current research work explores the potential of *Probabilistic Graphical Models* (PGMs), particularly Bayesian networks, to generate synthetic data representing novel cancer types through a model that includes existing cancer data and external environmental factors variation. The motivation behind this study is due to the limited dataset available on some cancers. It aims to discover new pathologies or forms of cancer evolution, accounting for the possibility of initiating gene rewriting in cancer.

In this research, we aim to generate new pathologies beyond those currently known. We utilize Bayesian learning to construct a model for the cancer clonal process and the subclonal process experimented with the environmental factor on the genetic structure of cancer. Using Bayesian model learning we generate cancer's protein mutation structure in a Directed Acyclic Graph with their respective conditional probability distributions (CPDs) allowing synthetic data sampling. The generated synthetic protein statistically corresponds to the original data.

The next step is to explore the generation of new pathologies from the Mixture Model using *Variational Autoencoders* (VAEs) and classify them against the known types of cancer.

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